Simulation-Based Analysis of Technician Dispatching Policies in a Field Service System

A home-appliance service company operates a field service system in Heilbronn. The company handles repair and installation requests at customer locations. All technicians start each working day at a central depot and return to the depot after their last attempted job. The requests for each day are known before route planning begins.

Service region and travel-time model

The service region is a 10×10 grid. A location is $c_i = (x_i, y_i)$, where x_i and y_i belong to $\{0,1,\ldots,9\}$. Customer locations are generated uniformly over the 100 grid intersections. The baseline depot is $c_0 = (5,5)$. The Manhattan distance between $c_i = (x_i, y_i)$ and $c_j = (x_j, y_j)$ is:

$$d_{ij} = |x_i - x_j| + |y_i - y_j|$$

A trip leg is one movement between two consecutive route locations. It can connect the depot to a customer, two customers, or a customer to the depot. Travel time is stochastic, and each trip includes an additional fixed preparation component of 5 minutes. For the trip leg from c_i to c_j, let $\tilde{T}_{ij}$ be the stochastic travel time per unit grid distance. The total trip-leg travel time is:

$$\tilde{T}_{ij} \sim \text{Triangular}(5,7.5,12), \qquad T_{ij} = 5 + d_{ij}\tilde{T}_{ij}$$

Route planning uses expected travel times. Since

$$\mathrm{E}[\tilde{T}_{ij}] = \frac{5 + 7.5 + 12}{3} = 8.1667,$$

The expected travel time for the trip leg is:

$$\mathrm{E}[T_{ij}] = 5 + 8.1667 d_{ij}$$

Daily shift, planning, and execution

Each working day contains one 8-hour shift, equal to 480 minutes. All technicians are available at the depot at the start of the day.

During planning, job j may be appended to technician k's route only if the remaining planned time covers:

- expected travel from the technician's current planned location c_i to customer location c_j,
- expected service time for job j when performed by technician k,
- expected return travel from c_j to the depot c_0, and
- a planning safety allowance of 20 minutes.

A candidate assignment is feasible when:

$$\mathrm{E}[T_{ij}] + \mathrm{E}[S_{jk}] + \mathrm{E}[T_{j0}] + 20 \le R_k$$

where, R_k is technician k's remaining planned time and $\mathrm{E}[S_{jk}]$ is the expected service time for job j performed by technician k. The additional 20-minute term is the planning safety allowance for travel-time and service-time variability. After a job is accepted, R_k is reduced by

$\mathrm{E}[T_{ij}] + \mathrm{E}[S_{jk}]$. The expected return time and the planning safety allowance remain reserved through the planning feasibility check.

During execution, each technician follows the planned route in sequence. Before each planned job, the model checks whether the remaining shift time covers the expected travel to the customer, the expected service, and the expected return travel. If the check fails, the job is classified as not attempted and carried over to the next day. A job classified as not attempted is not a failed repair attempt.

Actual travel and service times are sampled from their distributions. Because feasibility checks use expected times, a technician may finish after 480 minutes. This overtime is included in working time and utilization. A technician returns to the depot from the location of the last attempted job. A technician who attempts no job remains at the depot.

Customer demand, daily active pool limit, and priority

Let N_d be the ordered set of requests generated on day d. The daily number of generated requests follows:

$$|N_d| \sim \mathrm{Poisson}(80)$$

Each generated request is an installation with probability 0.60 and a repair with probability 0.40. A generated request becomes a job only after it is accepted into the daily active pool.

The daily active pool is limited to 160 jobs. Carry-over jobs enter first then the remaining active pool limit is filled with the new arrivals in job identifier order. The job identifier is a unique sequential number assigned to each request when it is generated and used to distinguish jobs and resolve deterministic ties. Let C_{d-1} be the carry-over set entering day d, and let A_d be the number of new requests accepted on day d.

$$A_d = \min\{|N_d|, \max(0, 160 - |C_{d-1}|)\}$$

Requests beyond the remaining capacity are classified as not accepted. They do not enter the assignment process or the carry-over set.

A new installation job has priority 0, and a new repair job has priority 1. The priority increases by 1 after each unresolved day, up to 5. A job remains at priority 5 until completion.

Technician groups, service times, and repair outcomes

The baseline workforce contains 3 Type R1 repair technicians, 3 Type R2 repair technicians, and 3 dedicated installation technicians. Type R1 and Type R2 technicians can perform repairs and installations. Dedicated installation technicians can perform installations only.

The service time of job j when performed by technician k is S_{jk}. Service times follow a gamma distribution with shape α and scale θ:

$$S_{jk} \sim \mathrm{Gamma}(\alpha, \theta), \qquad \mathrm{E}[S_{jk}] = \alpha\theta, \qquad \mathrm{SD}[S_{jk}] = \sqrt{\alpha}\theta$$

Job-technician combination	Gamma shape	Gamma scale
Installation by any technician	16	1.5
Repair by Type R1	4	5
Repair by Type R2	9	$5/3$

Table 1: Service-time parameters

A repair performed by a Type R2 technician always succeeds. A first-attempt repair performed by a Type R1 technician succeeds with probability 0.90. A failed Type R1 first-attempt repair becomes a second-attempt repair on the next day. Only a Type R2 technician may perform a second-attempt repair.

The end-of-day carry-over set contains:

- jobs not assigned during planning,
- planned jobs not attempted during execution, and
- failed Type R1 first-attempt repairs.

Dispatching policies

Both dispatching policies use the same first planning phase. They differ only in the assignment of jobs that remain after this phase.

Common first phase: dedicated installation technicians

Dedicated installation technicians are processed sequentially in technician identifier order. The planned route for each installation technician is determined before the next installation technician is considered.

Each technician starts at the depot and repeatedly selects the nearest feasible unassigned installation job. Distance from the technician's current planned location is the primary ranking criterion, and the job identifier is the deterministic tie-breaker. The jobs are checked in this order, and the first job satisfying the planning feasibility rule is assigned.

After each assignment, the technician's planned location and remaining planned time are updated. Jobs continue to be assigned to the same technician until no remaining installation jobs satisfy the planning feasibility rule or no jobs left. The procedure then continues with the next dedicated installation technician.

After this phase, remaining installation jobs and first-attempt repair jobs may be assigned to Type R1 or Type R2 technicians. Second-attempt repair jobs may be assigned only to Type R2 technicians.

Policy 1: Priority policy

The priority policy uses a job-first assignment rule. Jobs are selected and assigned individually. The policy does not completely fill one technician's route before considering other technicians.

At each assignment step, the highest-priority jobs are considered first. Among jobs with the same priority, jobs are ranked by their distance to the nearest eligible Type R1 or Type R2 technician from that technician's current planned location. Job identifier is the deterministic tie-breaker.

After selecting the highest-ranked job, the policy searches for a technician for that specific job. Eligible Type R2 technicians are considered before eligible Type R1 technicians. Within the same technician type, technicians are ranked by distance from their current planned location to the selected job, followed by technician identifier.

The job is assigned to the first eligible technician satisfying the planning feasibility rule. This means that Type R2 technicians are preferred for each individual job, but their complete routes are not necessarily filled before Type R1 technicians are considered.

After each accepted assignment, the selected technician's planned location and remaining planned time are updated. These updates may change the distances and feasibility conditions used during subsequent assignments.

If no eligible technician can accept the selected job, the job remains unassigned for the day. The policy continues with the remaining jobs until no remaining job can be assigned to any eligible technician.

Policy 2: Nearest policy

The nearest policy uses a technician-first greedy assignment rule. Type R2 technicians are processed first in technician identifier order, followed by Type R1 technicians. The planned route for each technician is constructed before the next technician is considered. Consequently, all Type R2 technician routes are constructed before any Type R1 technician route is constructed.

Each technician starts at the depot and repeatedly selects the nearest feasible eligible job. Distance from the technician's current planned location is the primary ranking criterion. Higher job priority is the first tie-breaker, and job identifier is the final deterministic tie-breaker.

The jobs are checked in this order, and the first job satisfying the planning feasibility rule is assigned. After each assignment, the technician's planned location and remaining planned time are updated.

Jobs continue to be assigned to the same technician until no eligible jobs remain that satisfy the planning feasibility rule. The procedure then continues with the next technician.

Job status and performance measures

Each generated request must have one final daily status: completed, not accepted, not assigned, not attempted, or failed first-attempt repair. Requests classified as not accepted are not carry-over jobs.

Let C_d be the carry-over set at the end of day d, let $p_{j,d}$ be the priority of job j at that time, and let D_d be the number of requests not accepted on day d. The company evaluates daily service performance using a composite penalty score:

$$Z_d = \sum_{j \in C_d} p_{j,d} + 5D_d$$

The score accounts for two sources of service shortfall. Carry-over jobs are penalized according to their end-of-day priority, while for each request that is not accepted because of the system limit, a penalty cost of 5 is added. A lower score, therefore, indicates better service performance. Average travel time is not included in this score and is reported separately.

The two dispatching policies are compared using the following key performance indicators:

- average daily service performance score;
- first-time fix rate for repair requests;
- probability that a repair request experiences a failed first service attempt;
- average daily travel time per technician type;
- technician utilization per technician type;
- workload balance, measured as average absolute deviation from the daily average utilization;
- average customer waiting time from request arrival to successful completion.

Tasks

Adapt the supplied python notebook **(YOU ARE NOT ALLOWED TO CHANGE THE EXISTING FUNCTIONS AND IMPLEMENTATION LOGIC.)** to answer the following tasks:

1. Determine an appropriate warm-up length and simulation horizon to obtain reliable estimates.
2. Report on the current performance of the system under both policies using the key performance indicators defined above.
3. Estimate the probability that a repair request experiences a failed first service attempt.
4. Conduct a 2^k factorial staffing experiment, where factors correspond to the technician groups. Each factor has two levels: the current staffing level and the current staffing level plus one additional technician in the corresponding group. Based on the results, analyze the main effects and the interaction effects.
5. Conduct a full factorial experiment with three levels for each of the following operational factors: daily request arrival rate, travel-time condition, and Type R1 repair success probability. Evaluate the resulting scenarios using the key performance indicators and compare the performance of the two policies across all factor combinations.
6. Analyze the spatial distributions of not assigned, not attempted, and failed first-attempt repair jobs under both policies. Explain and compare the spatial assignment patterns for both policies.
7. Compare average daily travel time per technician and technician service-location heat maps and explain the policy differences.
8. Analyze daily technician working times and workload fairness using the average absolute deviation from the daily average utilization.
9. Due to the enlargement of Bildungscampus, the central depot must be relocated to the grid position $c_0 = (3,3)$. Assess whether the baseline staffing structure remains adequate using the key performance metric for both policies.
10. Summarize the managerial implications and recommend a dispatching policy. State the conditions under which each policy is more appropriate.

Submission and Presentation

The submission deadline for Project 2 is July 20th at 11:59 a.m. You must submit your code (.ipynb) and presentation document (.pdf) on Moodle in the designated section. Make sure that you follow a specific approach to obtain your results.

The presentation should include the following:

- A brief introduction to the problem and the implemented policies.
- Explanations of the obtained results, answers to the tasks, and the methods used to obtain them for each task.
- Recommendations for the dispatching manager based on the simulation results.

Your code should include clear explanations of the implemented approaches through comments. You should not present any code during the presentation.

You will work in a group of up to 2 people. The presentation should be prepared for 20 minutes. If you work in a group, each member should be able to present all parts. During the presentations, a 10-minute section will be assigned to each person, followed by a 5-minute question round.

You should send your group information by 30.06.2026 to erdem.sahin@tum.de, including the names and student IDs of the group members. If you are unable to find a group member, contact us as soon as possible.

The presentations will be held on July 21st and 22nd. Preliminary time slots for the presentations are as follows:

21st July: 10:00–12:30, Room L.4.41
22nd July: 10:00–12:30, Room L.4.41

After the submission deadline, we will assign a 30-minute presentation slot to each group. Everyone must be present during their presentation slot. Further information will be provided after the allocation of slots.